Rule-Based Modeling of LDLR-Mediated LDL Uptake to Evaluate the Functional Impact of Coding Variants

Introduction

Familial hypercholesterolemia (FH) is a common genetic disorder characterized by elevated low-density lipoprotein (LDL) cholesterol and a substantially increased risk of premature cardiovascular disease (Brunham et al., 2018; Gidding et al., 2015). A major cause of FH is pathogenic variation in the LDL receptor (LDLR), the membrane protein responsible for binding circulating LDL particles and mediating their endocytosis into hepatocytes. Although thousands of LDLR coding variants have been identified, only a minority have experimentally defined functional effects.

Recently, a study from Tabet et al. (2025), *"The functional landscape of coding variation in the familial hypercholesterolemia gene LDLR,"* systematically quantified the activity of thousands of LDLR coding variants using high-throughput cellular assays. Their work provides an extensive map of how individual mutations influence LDL binding and receptor function, offering an important resource for interpreting genetic variation associated with FH. However, these experimental measurements primarily describe steady-state functional outcomes and do not capture the underlying dynamical processes—such as LDL binding kinetics, endocytosis, and recycling—that collectively shape LDL uptake over time.

To address this, my project aimed to develop a mechanistic model of LDLR-mediated LDL uptake and to test whether experimentally derived variant functional scores can drive meaningful differences in LDL uptake dynamics. Using a simplified rule-based modeling framework, I constructed a minimal representation of LDLR binding, internalization, and recycling that can be perturbed according to variant-specific parameters. The central question about this work is whether such a simplified dynamic model, parameterized directly by experimental functional scores, can reproduce the differential uptake patterns observed in cells. It provides insight into how coding mutations propagate through the LDL uptake pathway and establishes a foundation for building more comprehensive, detailed models in future studies.

Methods

Model Construction

A rule-based model of LDLR-mediated LDL uptake was developed using BioNetGen (Harris et al., 2016; Faeder, Blinov & Hlavacek, 2009). In the model, the LDLR was represented with four independent ligand-binding modules—LA3, LA4, LA5, and LA7—representing major LDL-binding domains. LDL particles were represented with a single receptor-binding site and three location states (extracellular, endosomal, and lysosomal). Trafficking events include internalization of LDLR–LDL complexes, release of LDL in the endosome, recycling of receptors back to the plasma membrane, and lysosomal degradation of LDL. The set of kinetic

parameters and initial conditions used in the model is provided in Table 1. Variant effects were incorporated using functional scores obtained from Tabet et al. (2025). For each variant, its experimental functional score (Table 2) was used to scale the surface dissociation rate (Table 1), while all other parameters remained fixed.

Model Assumptions

In our model, LDLR was restricted to binding a single LDL particle at any time, although its four LA modules contributed independently to the effective affinity. Extracellular LDL availability was maintained using catalytic-style binding rules that prevent ligand depletion, ensuring that binding events reflect receptor abundance rather than ligand limitation.

Simulation and Analysis

Deterministic simulations of the LDLR–LDL uptake model were performed using the ODE solver. Analysis and visualization of simulation output were conducted in Python using NumPy, pandas, and Matplotlib. Internalized LDL was quantified from the time-course trajectories as the sum of LDL in the endosomal and lysosomal compartments. Variant simulations were implemented by instantiating the model with the functional scores listed in Table 2, running the corresponding trajectories, and making a comparison. Two separate analysis scripts generated the validation plots and the pathogenic–benign separation analysis plot, including Pearson correlation and mean absolute error, to evaluate agreement between model-predicted and experimental functional scores.

Results and Discussion

Iterative Development of the LDLR–LDL Uptake Model

The LDLR–LDL uptake model progressed through three iterative refinements, each addressing inconsistencies between simulated behavior and known features of LDLR trafficking. The initial version (Figure 1a) displayed a major limitation that extracellular LDL was rapidly and almost depleted within the first few time steps. Because ligand molecules were consumed, LDLR–LDL interactions could not be sustained throughout the simulation, preventing meaningful differentiation of LDLR variants. This behavior indicated that the initial binding rules removed LDL from the extracellular pool in a manner inconsistent with physiological ligand abundance.

A second model revision (Figure **1b**) introduced one extra LDL molecule in the binding reaction to prevent ligand depletion and maintain a sustained extracellular LDL pool. While this modification successfully preserved extracellular LDL levels, it revealed a more severe structural artifact. Surface LDLR collapsed to zero shortly after simulation onset because the model permitted LDL to bind independently to each LA module, producing multivalent LDLR–LDL complexes. The emergence of these kinetically trapped complexes led to trafficking dynamics that differed substantially from those expected of a continuously recycling receptor system.

To address this issue, a final restructuring of the binding mechanism was implemented (Figure 1c). In this version, LDLR could bind at most one LDL particle at a time, while individual LA

modules continued to contribute to the effective affinity. Enforcing single-ligand occupancy eliminated the formation of multivalent complexes and restored realistic receptor cycling.The corrected model produced stable surface and endosomal LDLR pools, maintained physiologically reasonable extracellular LDL levels, and generated coherent LDL uptake trajectories. This refined formulation provided a reliable foundation for downstream variant simulations and for investigating the model's sensitivity to variation in binding affinity.

Variant Simulations Reveal Limited Dynamic Range and Poor Agreement with Experiment

To evaluate whether the model could reproduce experimental measurements of variant function, LDL uptake was simulated for five LDLR variants with functional scores reported by Tabet et al. (2025) (Table 2). They included wild type, two pathogenic variants and two benign variants. Based on the experimental data, pathogenic variants with markedly reduced activity were expected to yield substantially lower LDL uptake compared with wild type and benign variants.

However, the model failed to reflect these differences. As shown in Figure 2, the model-predicted uptake values for all five variants were nearly identical, clustering tightly between 0.95 and 1.00 after normalization to wild type. Pathogenic variants showed essentially the same uptake as benign variants, and the resulting correlation between experimental and model scores was weak ($r = 0.458$, $p = 0.438$). In addition, a complementary analysis (Figure 3) showed no separation between pathogenic and benign groups, with neither group differing appreciably from wild type. These observations indicate that, in its current form, the model is unable to propagate variant-dependent changes in binding affinity into measurable differences in LDL uptake.

Sensitivity Analysis Shows LDL Uptake Is Weakly Affected by Changes in LA3 Binding Strength

To understand why variant effects failed to influence system-level behavior, a parameter scan was performed over a 100-fold range of LA3 binding strengths. This parameter directly contributes to the dissociation rate. Across this wide range of perturbations, total LDL uptake varied by less than 0.2% (Figure 4), indicating that the simulated uptake dynamics are only minimally sensitive to changes in binding strength.

To investigate the mechanistic basis for this insensitivity, time-course trajectories were examined (Figure 1c). LDLR–LDL complexes were internalized shortly after their formation, leaving limited opportunity for dissociation to occur. Because very few unbinding events took place before internalization, alterations in dissociation rate had little effect on the amount of LDL that ultimately reached the lysosomal compartment.

As a result, the system behaved similarly across the entire parameter range, with all simulations converging to nearly identical uptake values. These findings explain why variant-dependent

changes in binding affinity failed to propagate into measurable differences in LDL uptake in the earlier simulations.

Model Limitations and Directions for Future Development

The results of the variant simulations and sensitivity analysis reveal several structural limitations in the current LDLR–LDL uptake model.

First, the endocytosis rate is faster than expected, allowing LDLR–LDL complexes to be internalized almost immediately after formation. Because unbinding events rarely occur before internalization, differences in receptor–ligand affinity cannot influence LDL uptake. As a result, endocytosis, rather than binding kinetics, determines the fate of LDL in the model. A second limitation is that binding occurs much more rapidly than unbinding, meaning that receptor–ligand interactions reach a steady state almost immediately. Under this condition, changes in binding affinity have very limited influence on overall LDL uptake. Third, the structural representation of LDLR is highly simplified. Only four LA modules are included, whereas real LDLR variants can affect the entire LA1–LA7 region, the EGF-A and β-propeller domains, and the cytoplasmic tail that governs trafficking and recycling (Tabet et al., 2025). Many pathogenic variants primarily impair LDLR maturation rather than binding affinity, which is not represented in this first-version model. In addition, in our model, extracellular LDL is treated as a buffered pool through catalytic binding rules, preventing ligand depletion or competition from shaping uptake dynamics.

These limitations point directly toward future improvements. Rebalancing kinetic parameters, such as adjusting binding rates or slowing endocytosis, may allow LDLR variants to have a measurable impact on LDL uptake. Incorporating cooperative interactions among LA modules would allow more detailed representation of domain-specific contributions to binding. Module-specific analyses could then determine which domains have the greatest influence on uptake dynamics. Finally, expanding the model to include VLDL binding would allow examination of LDLR function in a broader physiological context, where multiple lipoproteins compete for receptor occupancy. Collectively, these refinements would allow future versions of the model to capture a richer set of variant effects and more closely match experimentally observed behavior.

References

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Table 1. Parameters used in the rule-based model of LDLR-mediated LDL uptake.

Parameter	Description	Value

k_{on}	Basal LDL association rate at the plasma membrane	1.0
$k_{off,surf}$	Variant-dependent LDL dissociation rate at the surface	1/max(s,0.01)
$k_{off,endo}$	LDL dissociation rate in endosomal compartment	50.0
$strength_{LA3}$	Effective affinity contribution of LA3 domain	1.0
$strength_{LA4}$	Effective affinity contribution of LA4 domain	1.0
$strength_{LA5}$	Effective affinity contribution of LA5 domain	1.0
$strength_{LA7}$	Effective affinity contribution of LA7 domain	0.8
k_{endo}	Endocytosis rate of LDLR–LDL complexes	2.0
$k_{recycle}$	Recycling rate of unbound LDLR from endosome to surface	3.0
$k_{degrade}$	Rate of LDL progression toward lysosomal degradation	5.0
$LDLR_{init}$	Initial number of surface LDLR molecules	1000
LDL_{conc}	Initial number of extracellular LDL particles	100

Table 2. Experimental functional scores for the five LDLR variants simulated in this study.

Variant	Functional Score	Clinical Classification
WT	1.00	Wild type
C52Y	0.05	Pathogenic
D147N	0.15	Pathogenic
P526L	0.95	Benign
T705I	0.92	Benign

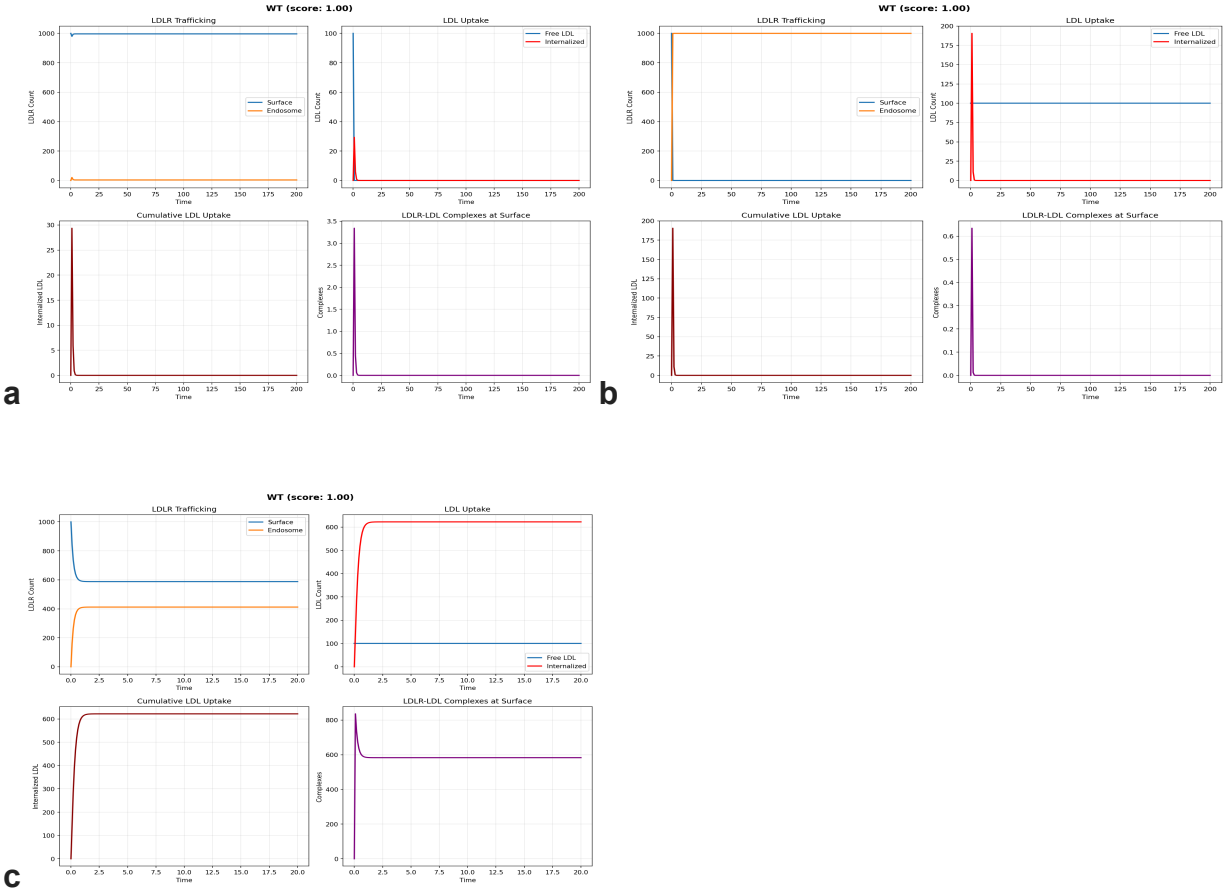

Figure 1. Iterative refinement of the LDLR–LDL uptake model.
(a) First version wild-type LDLR–LDL uptake model showing four outputs: surface and endosomal LDLR trafficking, LDL uptake dynamics, cumulative internalized LDL, and surface LDLR–LDL complex formation.
(b) Second version wild-type LDLR–LDL uptake model with the same four outputs.
(c) Final version wild-type LDLR–LDL uptake model with the same four outputs.

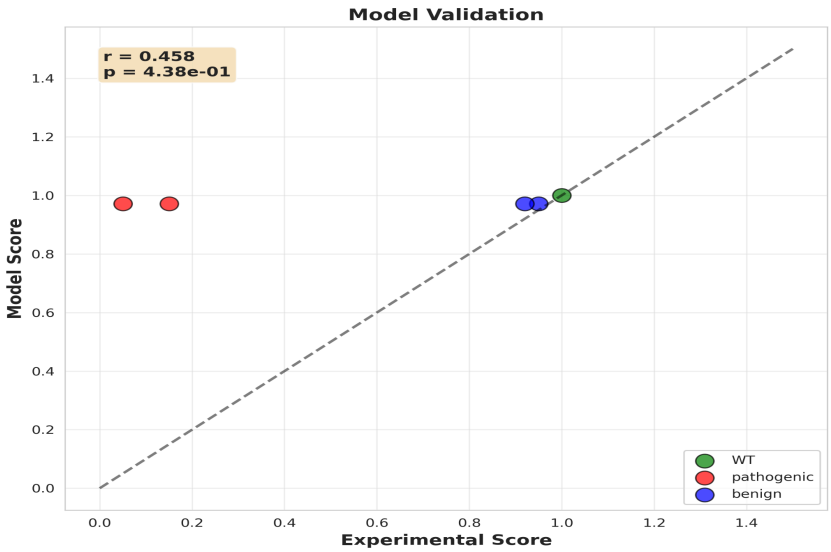

Figure 2. Model-predicted functional scores compared with experimental measurements. Scatter plot showing experimental functional scores (x-axis) from Tabet et al. (2025) versus model-predicted LDL uptake (y-axis), normalized to wild type. Colors indicate variant class (green: wild type; red: pathogenic; blue: benign). All simulated variants cluster between 0.97 and 1.00, resulting in poor correlation with experiment (r = 0.458, p = 0.438). The dashed line indicates the identity line, indicating perfect agreement.

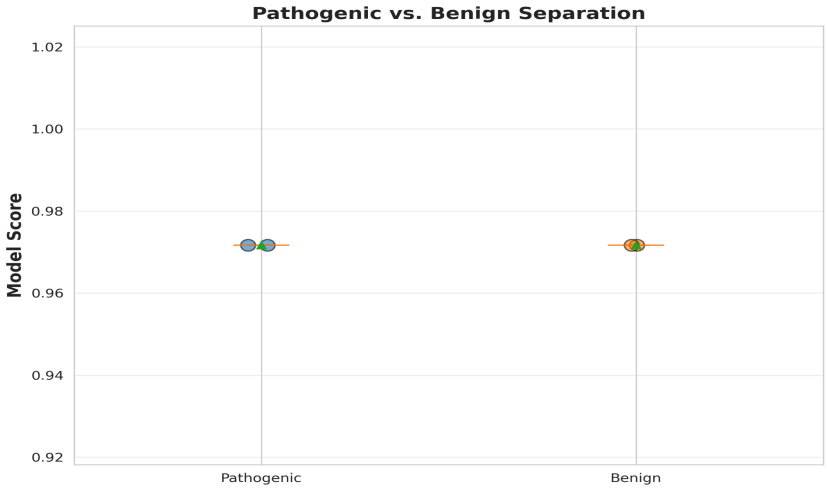

Figure 3. Separation between pathogenic and benign variants in model simulations. Model-predicted LDL uptake values for pathogenic and benign LDLR variants. Uptake values are grouped by clinical classification to display the distribution of model-predicted functional scores across variant classes.

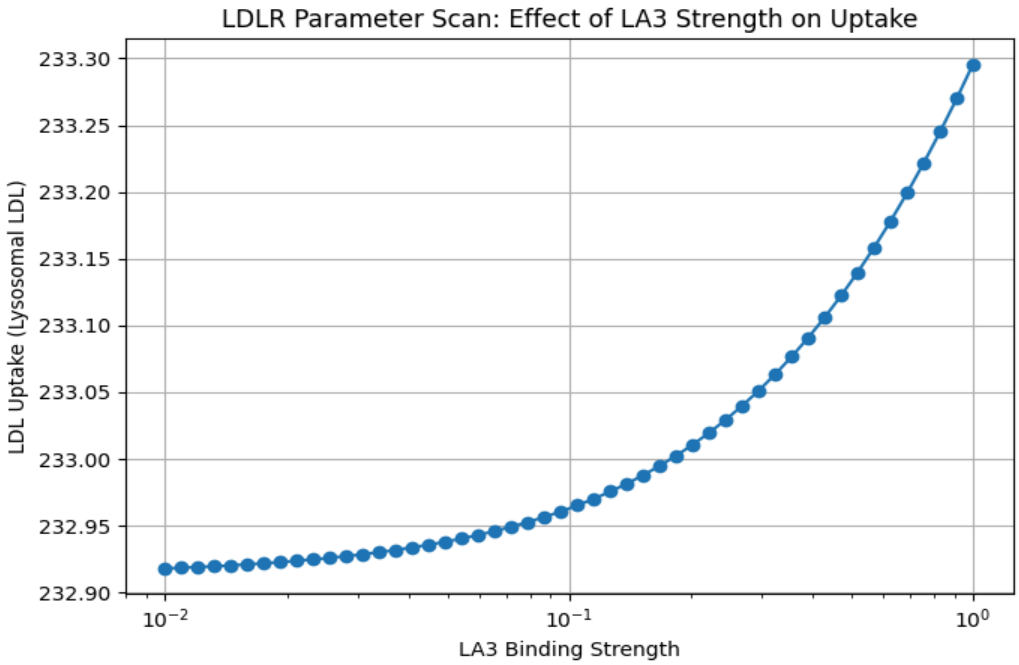

Figure 4. Sensitivity of LDL uptake in lysosome to variation in LA3 binding strength. Lysosomal LDL levels plotted as a function of LA3 binding strength, varied over a 100-fold range on a logarithmic scale. The curve displays the relationship between changes in the LA3 affinity parameter and total LDL uptake at the final simulation time point.